

Figure 4: Go to the path of CloudSim library

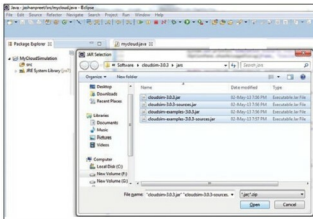


Figure 5: Select all JAR files of CloudSim for integration

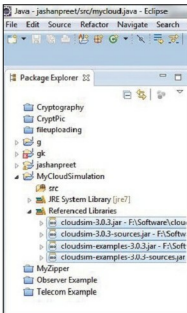


Figure 6: JAR files of CloudSim visible in the referenced libraries of Eclipse with Java Project

3. Give a name to your project.
4. Configure the build path for adding the CloudSim library.
5. Search and select the CloudSim JAR files.

In the integration and implementation of Java code and CloudSim, the Java-based methods and packages can be used. In this approach, the Java library is directly associated with CloudSim code.

After executing the code in Eclipse, the following output will be generated, which makes it evident that the integration of the dynamic key exchange is implemented with the CloudSim code:

Starting Cloud Simulation with Dynamic and Hybrid Secured Key Initialising...

MDS Hash Digest(in Hex. format)::

6e47ed3d3cde35ef1cc198a78d3da9c9f

Hybrid Approach (SHA+MD5) Hash Hex format:

b0a309c58489d6788262859da2e7d45b6ac20a052b6e606ed1759648e43e40b

Hybrid Approach Based (SHA+MD5) Security Key Transmitted =>

ygcsbyybr4c ^cE7j@E

Starting CloudSim version 3.0

CloudDatacentre-1 is starting...

CloudDatacentre-2 is starting...

Broker is starting...

Entities started.

0.0: Broker: Cloud Resource List received with 2 resource(s)

0.0: Broker: Trying to Create VM #0 in CloudDatacentre-1

0.0: Broker: Trying to Create VM #1 in CloudDatacentre-1

[VmScheduler.vmCreate] Allocation of VM #1 to Host #0 failed by MIPS

0.1: Broker: VM #0 has been created in Datacentre #2, Host #0

0.1: Broker: Creation of VM #1 failed in Datacentre #2

0.1: Broker: Trying to Create VM #1 in CloudDatacentre-2

0.2: Broker: VM #1 has been created in Datacentre #3, Host #0

0.2: Broker: Sending cloudlet 0 to VM #0

0.2: Broker: Sending cloudlet 1 to VM #1

0.2: Broker: Sending cloudlet 2 to VM #0

160.2: Broker: Cloudlet 1 received

320.2: Broker: Cloudlet 0 received

320.2: Broker: Cloudlet 2 received

320.2: Broker: All Cloudlets executed. Finishing...

320.2: Broker: Destroying VM #0

320.2: Broker: Destroying VM #1

Broker is shutting down...

Simulation: No more future events

CloudInformationService: Notify all CloudSim entities for shutting down.

CloudDatacentre-1 is shutting down...

CloudDatacentre-2 is shutting down...

Broker is shutting down...

Simulation completed.

Simulation completed.

===== OUTPUT =====

Cloudlet ID	STATUS	Data centre ID	VM ID	Time	Start Time
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Finish Time

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1	SUCCESS	3	1	160	0.2	160.2
0	SUCCESS	2	0	320	0.2	320.2
2	SUCCESS	2	0	320	0.2	320.2

Cloud Simulation Finish

Simulation Scenario Finish with Successful Matching of the Keys
Simulation Scenario Execution Time in MillSeconds => 5767

Security Parameter => 30.95937277933122

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